EE5907 Pattern Recognition / EE5027 Statistical PR

Lecture 1 Foundations of Pattern Recognition and Machine Learning

© Copyright National University of Singapore. All Rights Reserved.

Welcome to EE5907!

- Course will be conducted by Dr. XU Yuecong (1st part, till Recess) and Dr. Wang Si (2nd part, Wk 7 – 13).
- Topics on decision theory, linear/non-linear classifier and regressor, PCA, LDA, clustering, boosting, SVM, etc.
- Syllabus available on Canvas in the Pages tab, please check them out.

© Copyright National University of Singapore. All Rights Reserved.

Lecture Format and Consultation

- In-person lectures:
  - Lecture notes will be available right before each class.
  - Recording will be available at the end of each class for revision.
  - Break for 15 mins in between, end in 2.5 hours. After class discussion welcomed!
- Consultation hour: walk-in if you have any questions:
  - Mon: 14:00 – 15:30, Week 2, 5, 6. ECE Conference Room E4-05-39
  - Mon: 13:00 – 14:30, Week 3 (25 Aug). ECE Conference Room
  - Announcements will be made if there are changes for that week.
- Q&A: https://docs.google.com/document/d/1ph1hVnzrmUsk0-fTsiIF5OPGm2wHQQw4qrpa-pQa5Ag/edit?usp=sharing

© Copyright National University of Singapore. All Rights Reserved.

Lecturers

- Lecturers:
  - Dr. XU Yuecong (Part I, Week 1 – 6) yc.xu@nus.edu.sg
  - Dr. WANG Si (Part II, Week 7 – 13)
- Graduate Assistants (GA):

© Copyright National University of Singapore. All Rights Reserved.

Background of Myself and Course

- Video, 3D data and time-series data analysis
- Transfer learning for efficiency and application.
- Combining with LLVM and AIOT devices.

NUS National University of Singapore
Department of Electrical & Computer Engineering

yc.xu@nus.edu.sg

- Pattern recognition and machine learning, not deep learning.
- Understanding the concepts for application.
- Some DL algorithms/techniques are based on PR/ML.

© Copyright National University of Singapore. All Rights Reserved.

Learning Outcomes

- Comprehend principles and core techniques in pattern recognition such as Bayesian decision theory, linear and non-linear classifiers, principal component analysis, linear discriminant analysis, clustering and support vector machine.
- Apply the conceptual and mathematical tools of pattern recognition to analyze real-world data, obtaining data features and performing classification or regression.
- Assess the strengths and weaknesses of the different pattern recognition algorithms.

© Copyright National University of Singapore. All Rights Reserved.

Assessment and Grading

- Continuous Assessment consists of 3 components.
  - CA1: 29 Aug 2025 – 3 Oct 2025 SGT 23:59 (Week 3 – 7): 15%.
  - CA1 To be announced on Week 3.
  - Quiz: 18 Sep 2025 (During lecture on Week 6): 5%.
  - Quiz format: MCQ, fully open book, max 60 minutes.
  - Quiz to be conducted via Canvas.
  - CA2: TBA by Dr. Wang Si: 20%.

- Final: 60%.
  - Closed book with one-page double-sided cheat sheet.

© Copyright National University of Singapore. All Rights Reserved.

Assessment and Grading

- **Deadlines are strict.** Late submission will be deducted 10 points (out of 100) for every 24 hours. **ZERO** mark for late submission of 5 days (120 hours) or more.
- **NO plagiarism is allowed.** A **ZERO** mark will be given to anyone found plagiarizing and a formal report will be handed to the Department for further investigation.
- GPT-based/GPT-like tools (ChatGPT, Co-pilot, Gemini) allowed for refinement and coding (of course, amendments are required). They're **NOT allowed** for generating the whole project submission.

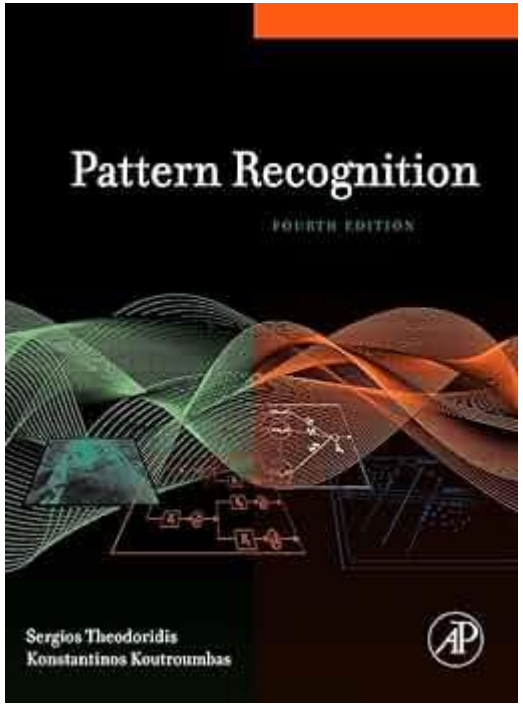
© Copyright National University of Singapore. All Rights Reserved.

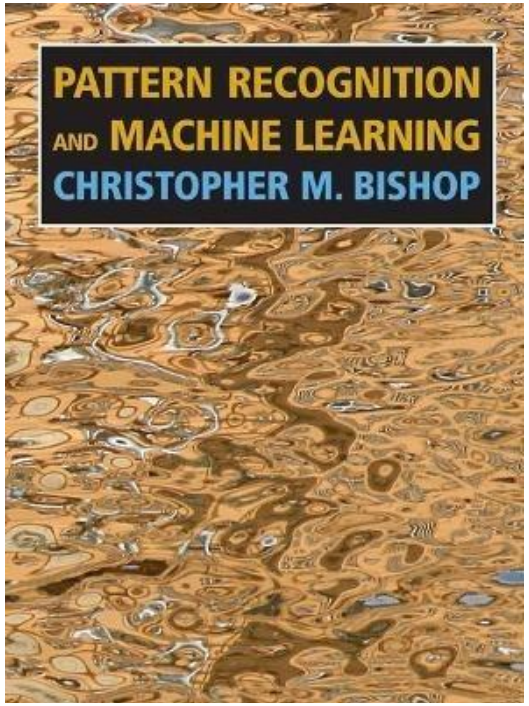
References

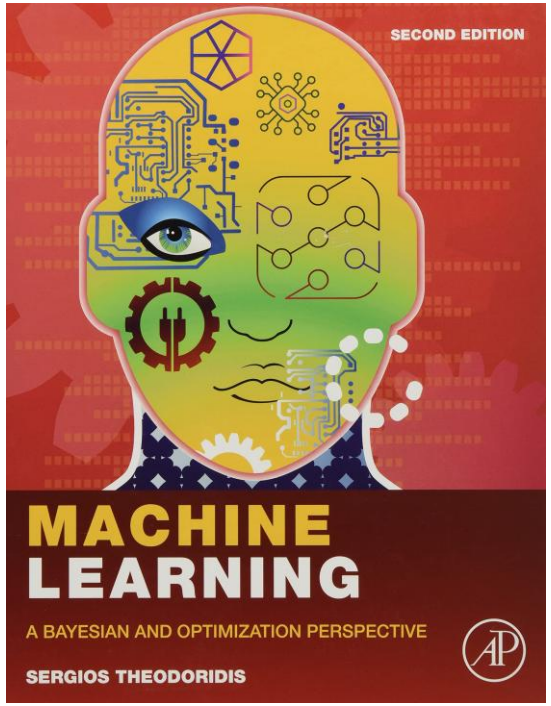
- Theodoridis, Sergios, and Konstantinos Koutroumbas. Pattern recognition. Elsevier, 2006.
- Bishop, Christopher M., and Nasser M. Nasrabadi. Pattern recognition and machine learning. Vol. 4, no. 4. New York: springer, 2006.
- Theodoridis, Sergios. Machine Learning: A Bayesian and Optimization Perspective. Academic Press, 2020.
- Murphy, Kevin P. Probabilistic machine learning: an introduction. MIT press, 2022.

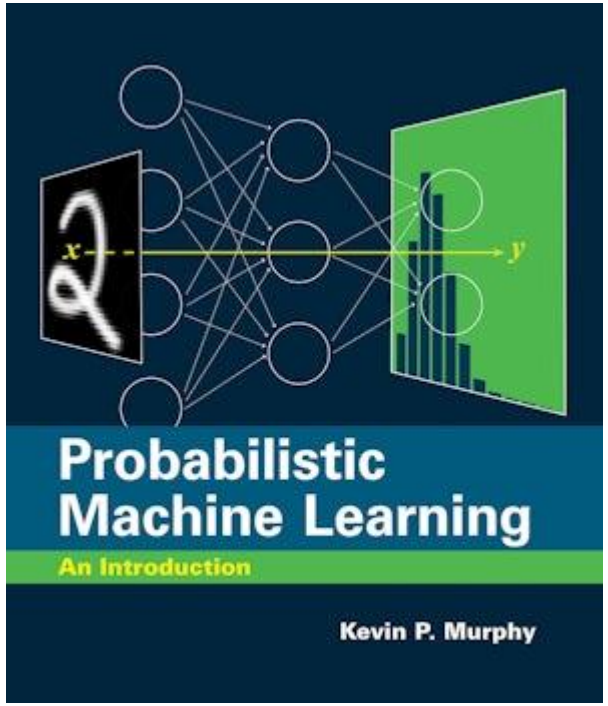
© Copyright National University of Singapore. All Rights Reserved.

References









© Copyright National University of Singapore. All Rights Reserved.

Code of Conduct

- Students are encouraged to actively participate.
  - Your grade depends on what you do, not what others do.
  - Don’t share your solutions/codes. Submit only your own work.
- Academic Integrity:
  - Student’s academic work & behavior are held to highest academic integrity standards. Cheating, plagiarism, unauthorized collaboration, and ‘helping others commit these acts’ are examples of academic misconduct, which can result in disciplinary action.

- **Any violation to academic honesty will be reported to the University's disciplinary panel.**

© Copyright National University of Singapore. All Rights Reserved.

Updated Syllabus (1st part) (1/2)

- Foundations of pattern recognition and machine learning:
  - Basic terminologies and concepts of PRML;
  - Bayes' rule and univariate probability distribution models;
  - Maximum Likelihood Estimation (MLE)
  - Empirical Risk Minimization (ERM).
- Decision theory and classifiers:
  - Bayesian decision theory and the classification problem;
  - Bayesian classifiers for normal distribution and Maximum A Posteriori (MAP);
  - The frequentist decision theory and its application.

© Copyright National University of Singapore. All Rights Reserved.

Updated Syllabus (1st part) (2/2)

- Linear classifiers and non-linear classifiers:
  - The perceptron algorithm and LSE for optimal linear classifiers;
  - Analysis of the XOR problem and the Multi-Layer Perceptron (MLP);
  - Training of MLPs and the Radial Basis Function Networks (RBF).
- Linear Regression and Non-parametric PR methods:
  - LSE for linear regression, the ridge regression and Lasso regression;
  - Classification And Regression Trees (CART);
  - Bootstrap aggregating and Random Forests (RF).

© Copyright National University of Singapore. All Rights Reserved.

Something More and New:

- Tutorials will be provided to help you understand the concepts better (and to better prepare for Finals ☺).
  - Not graded at all but will be relevant to the finals in the type of questions and the concepts that may be examined.
  - May not fully overlap but would be relevant.
  - Tutorials provided on Weeks 3, after Lecture Set 2.
  - Tutorials discussed on Weeks 4, during class.

© Copyright National University of Singapore. All Rights Reserved.

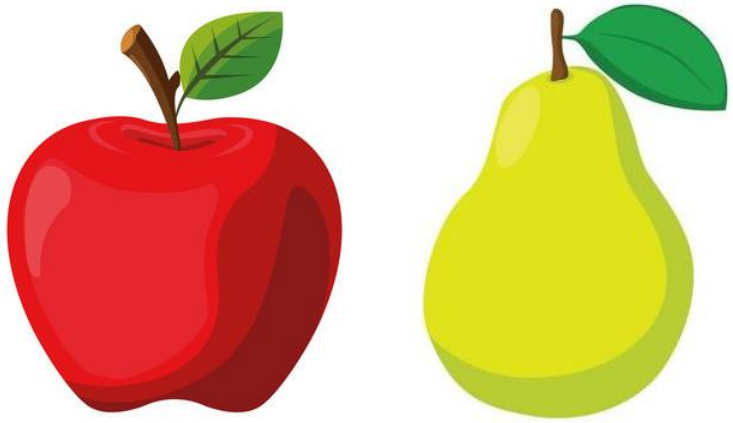
Introduction:

- ✓ What is Pattern Recognition / Machine Learning?
- ✓ Basic terminologies that may be encountered.

© Copyright National University of Singapore. All Rights Reserved.

What is Pattern Recognition (PR)?

- Pattern recognition is to **perceive** a pattern, **extract** the relevant info, **understand** the content of the info and **make decision** automatically by machine or computer.
  - In short: automated recognition of patterns and regularities in data.
  - Uses techniques developed in statistics and machine learning.
- Example 1: Is the picture of an apple or pear



- After perceiving the image, you may decide left image is an apple, right image is a pear.
- The decision is made with 100% certainty (probability): how?
- Based on color/shape.

© Copyright National University of Singapore. All Rights Reserved.

What is Pattern Recognition?

- A more complex Example 2: Is the picture of an apple or pear

- After perceiving the image, the decision is made only after extracting the item from the correct part of the image.
- If item is correctly extracted: decision can be made with 100% certainty.

© Copyright National University of Singapore. All Rights Reserved.

What is Pattern Recognition?

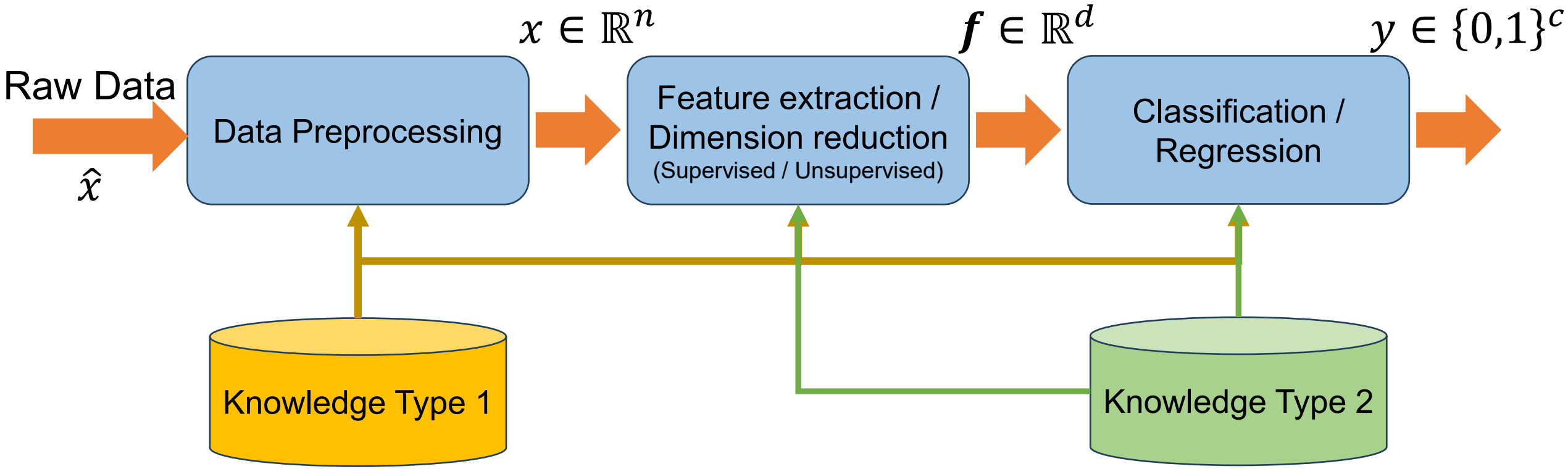
- Can we always make a decision with 100% certainty?
  - Example 3: Is the picture of an apple or pear

- Lost of certain information / features (color):
- Can we still make the decision with 100% certainty? Probably not – then how should we determine what the picture actually is?

© Copyright National University of Singapore. All Rights Reserved.

The PR System Diagram

What we usually get here:
01 → apple; 10 → pear;
What do 0 & 1 represent?
Probability!



© Copyright National University of Singapore. All Rights Reserved.

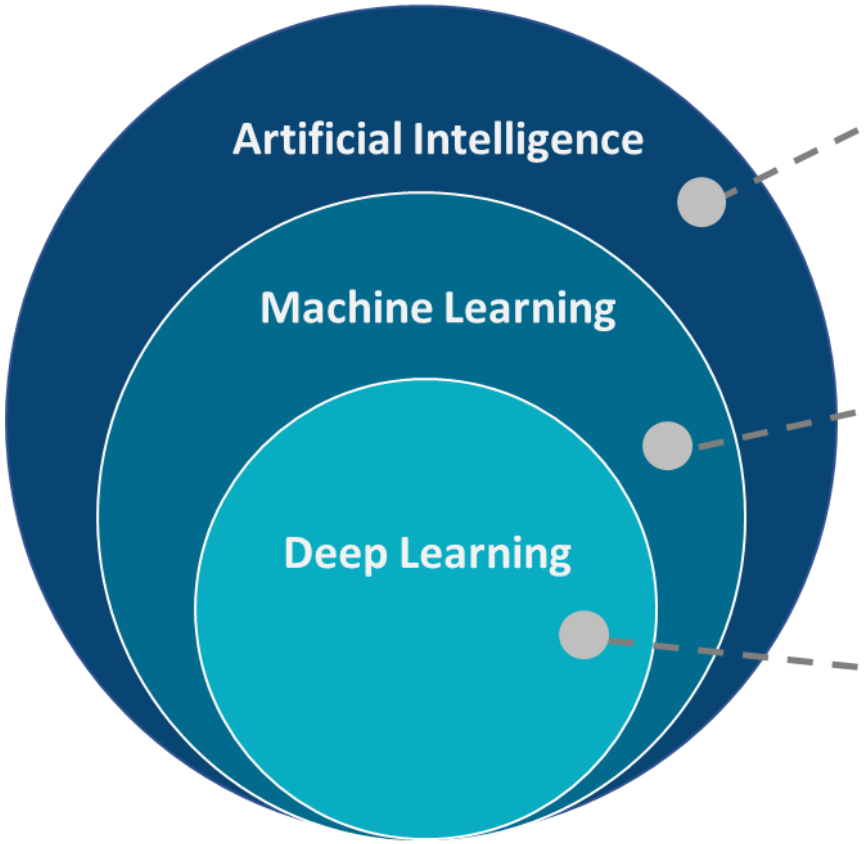
Pattern Recognition (PR) vs. / and Machine Learning (ML)

- Connections or differences between PR and ML?
  - Machine learning: learn from data without being explicitly programmed.
  - Almost identical to PR, since learning implies finding patterns.

- What is ML? Compare with non-learning algo (why need ML)?
  - Non-learning algos are bared on pre-determined rules, require human expertise. The rules are fixed regardless the input data.
  - For non-learning algos, human expert should know the solution and rules – problems that human cannot solve would NOT be solved.
  - Non-learning algo: human-centric; ML/PR: data-centric.

© Copyright National University of Singapore. All Rights Reserved.

The different concepts: AI / ML / DL



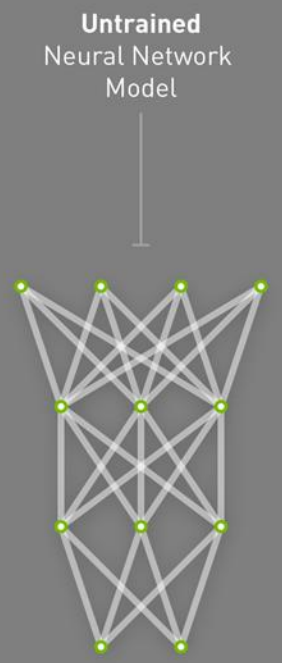
Artificial Intelligence
A technique which enables machines to mimic human behaviour

Machine Learning
Subset of AI technique which uses statistical methods to enable machines to improve with experience.

Deep Learning
Subset of Machine Learning which make the computation of multi-layer neural network feasible

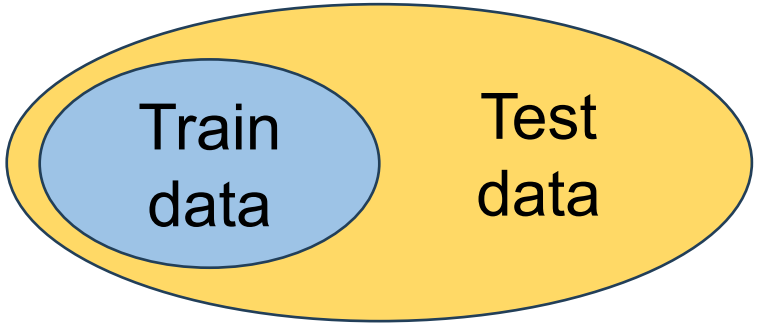
© Copyright National University of Singapore. All Rights Reserved.

The Common ML Framework



Inferencing: Deductive / Inductive Reasoning

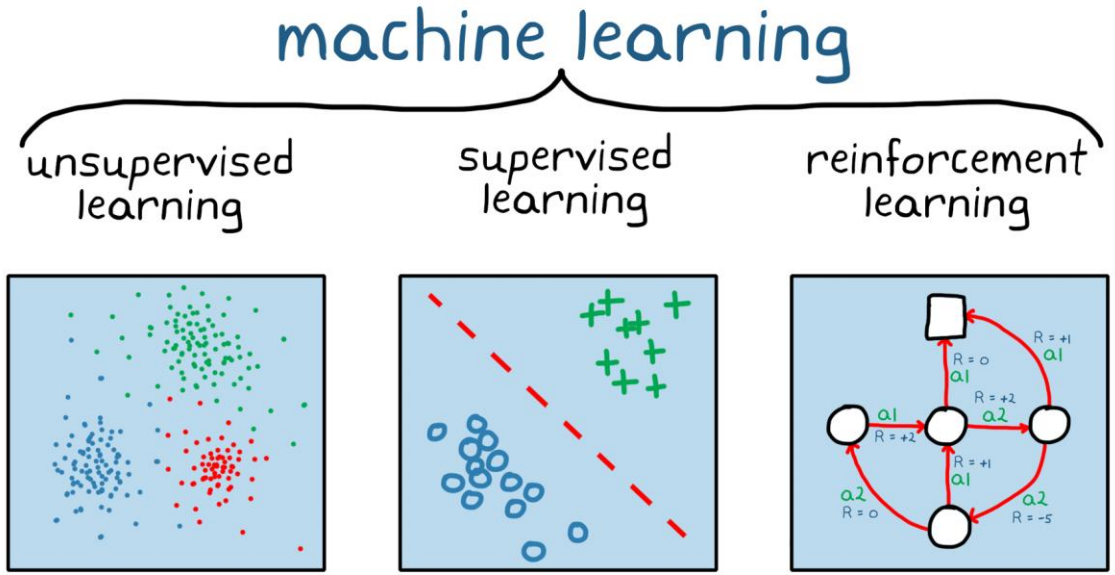
- Deductive reasoning is a logical approach where you go from **general ideas** to **specific conclusions**.
  - All men are mortal;
  - Socrates is a man;
  - → Socrates is mortal.
- Inductive reasoning is a method of drawing conclusion from the **specific** to the **general**.
  - All swans we've seen is white;
  - → All swans are white (?)
- Generalization



© Copyright National University of Singapore. All Rights Reserved.

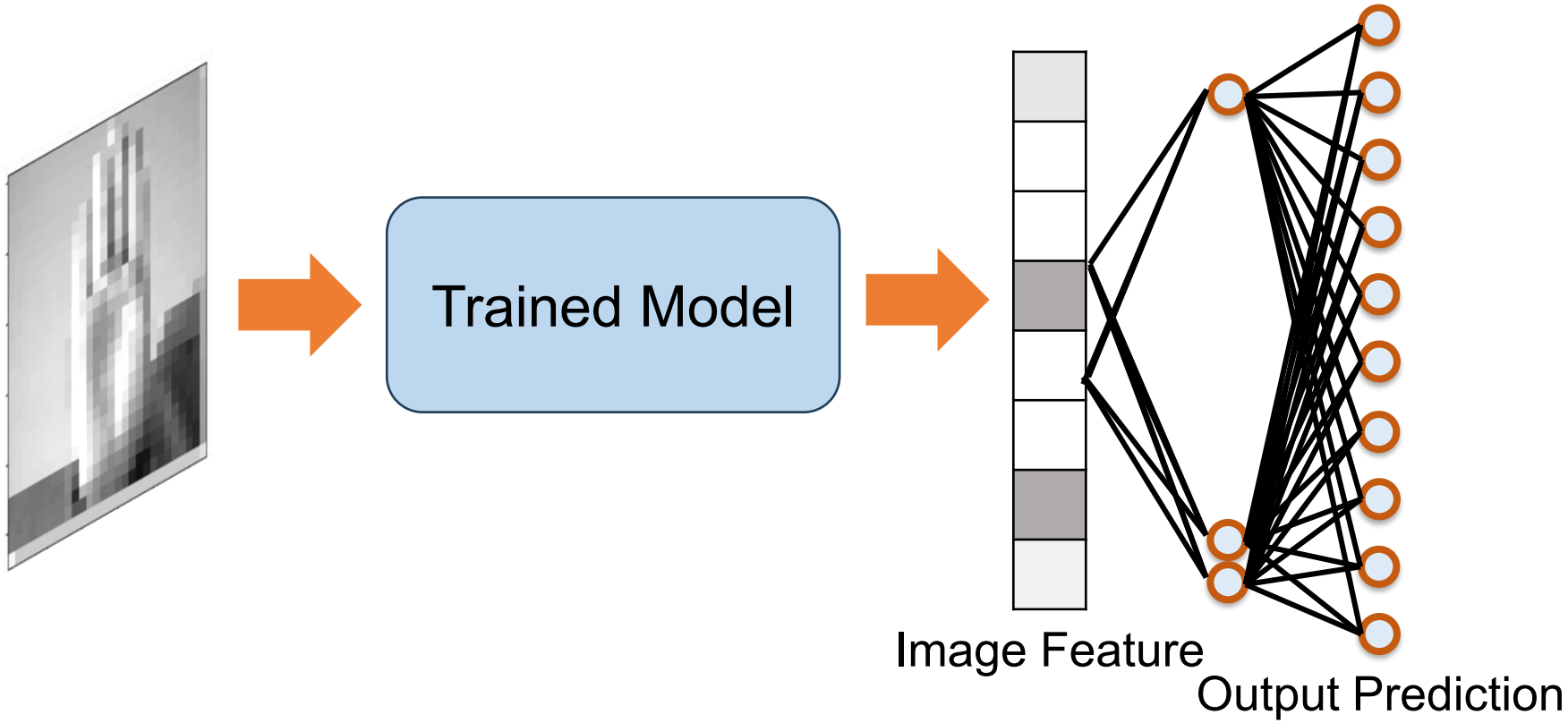
Different Learning Paradigms

- Supervised learning: Labelled data with guidance;
- Unsupervised learning: No labelled without guidance;
- Reinforcement learning: Interacts with environment, decide action, learns by trial-and-error method.



© Copyright National University of Singapore. All Rights Reserved.

The Common ML Framework: Output



- Note what is the usual output (for classification)?
- The **probability** of each class!

© Copyright National University of Singapore. All Rights Reserved.

Foundation

- ✓ Introduction to Probability.
- ✓ The Bayes' Rule.
- ✓ Univariate Probability Distribution Models.

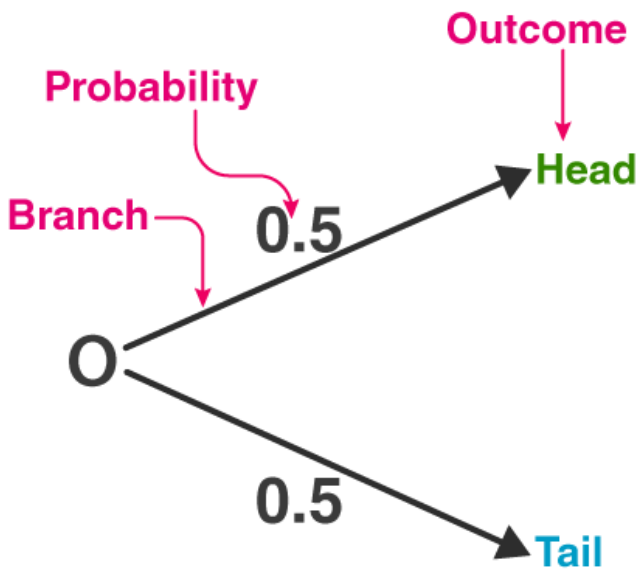
© Copyright National University of Singapore. All Rights Reserved.

Probability: Two Interpretations

- Probability theory is nothing, but common sense reduced to calculation.
  – Pierre Laplace, 1812

- Frequentist interpretation: probabilities represent long run frequencies of events that can happen multiple times.
  - Flip the coin many times, we expect it to land heads about half the time.

- Bayesian interpretation: quantify **uncertainty** or ignorance about something
  - The coin is equally likely to land heads/tails on next toss.



© Copyright National University of Singapore. All Rights Reserved.

Properties of Probability of event(s): I

- Note: for this course, probability of an event denoted as: $P(A)$

- $P(A)$ satisfy: $0 \leq P(A) \leq 1$.
- Joint probability that events A and B both happening: $P(A, B)$;
- If A and B are independent events, $P(A, B) = P(A)P(B)$.
- Probability of event A or B happening (probability of union of A and B): $P(A \vee B) = P(A) + P(B) - P(A, B)$;
- If A and B are mutually exclusive (cannot happen at the same time), then $P(A \vee B) = P(A) + P(B)$;

© Copyright National University of Singapore. All Rights Reserved.

Properties of Probability of event(s): II

- Note: for this course, probability of an event denoted as: $P(A)$ or $p(A)$.

- Conditional probability of B happening given A has occurred:

$$P(B|A) \triangleq \frac{P(A,B)}{P(A)}$$

- Event A is independent of B if $P(A,B) = P(A)P(B)$;
- Events A ad B are conditionally independent given C if:

$$P(A,B|C) = P(A|C)P(B|C)$$

© Copyright National University of Singapore. All Rights Reserved.

Discrete and Continuous Random Variables (RV)

- A mathematical formalization of a quantity or object which depends on random events.
- A mapping or a function from possible outcomes in a sample space to a measurable space, often the real numbers.

Discrete variable	Continuous variable
Countable numbers of value (e.g., number of students)	Infinitely many real values (e.g., time to complete a task)
Distribution defined by probability mass function (pmf)	Distribution defined by cumulative distribution function (cdf) and probability density function (pdf)
$p(x) \triangleq P(X = x)$ $\sum p(x) = 1$	$P(x) \triangleq P(X \leq x)$
	$p(x) \triangleq \frac{d}{dx} P(x)$ $P(a < X \leq b) = \int_a^b p(x)dx = P(b) - P(a)$

© Copyright National University of Singapore. All Rights Reserved.

Discrete and Continuous Random Variables (RV)

Discrete variable	Continuous variable
Countable numbers of value (e.g., number of students)	Infinitely many real values (e.g., time to complete a task)
Distribution defined by probability mass function (pmf)	Distribution defined by cumulative distribution function (cdf) and probability density function (pdf)
$p(x) \triangleq P(X = x)$ $\sum p(x) = 1$	$P(x) \triangleq P(X \leq x)$ $p(x) \triangleq \frac{d}{dx} P(x)$ $P(a < X \leq b) = \int_a^b p(x)dx = P(b) - P(a)$

- Important: Note whether the variable is discrete or continuous! Could be confusing sometimes. Example:
  - A large quantity of green tea (e.g., 1000kg) is to be sold. X is the weight of green tea to be sold. Suppose any quantity can be sold, is X discrete or continuous?
  - What if the green tea can only be sold in packets of 10kg, is X now discrete or continuous?

© Copyright National University of Singapore. All Rights Reserved.

Bayes' Rule

- "Bayesian": refer to inference methods that represent "degrees of certainty" using probability theory, and which leverage Bayes' rule, to update the degree of certainty given data.
  - *Layman term: Update our belief based on prior belief and data so that we infer something with a certain degree of uncertainty*.
- Bayes' rule: a formula for computing the probability distribution over possible values of an unknown quantity H given observed data $Y = y$.

$$p(H = h|Y = y) = \frac{p(H = h)p(Y = y|H = h)}{p(Y = y)}$$

- Follows the product rule of probability: $p(h|y)p(y) = p(h)p(y|h) = p(h, y)$
- $p(H = h|Y = y)$: **posterior distribution / conditional distribution**.

© Copyright National University of Singapore. All Rights Reserved.

Bayes' Rule: in Detail

$$p(H = h|Y = y) = \frac{p(H = h)p(Y = y|H = h)}{p(Y = y)}$$

- $p(H)$: what we know about possible values of H before we see any data – the **prior distribution** (layman terms: prior belief).
- $p(Y|H = h)$: the distribution over possible outcomes Y we expect to see if $H = h$ – the **likelihood** (layman terms: the *observed* data).

- Multiplying the prior distribution and the likelihood – unnormalized joint distribution $p(H = h, Y = y)$. Convert into a normalized distribution by dividing by $p(Y = y)$, which is denoted as the **marginal likelihood**.
- $p(H = h|Y = y)$: represents our new **belief state** (updated belief) about the possible values of H – the **posterior distribution**.

$$\text{posterior} \propto \text{prior} \times \text{likelihood}$$

© Copyright National University of Singapore. All Rights Reserved.

Bayes' Rule: Example 1 – Weather Forecast

- Marie is getting married tomorrow at an outdoor ceremony in the desert.
- In recent years, it has rained only 5 days each year.
- When it actually rains, the weatherman has forecast rain 90% of the time.
- When it doesn't rain, he has forecast rain 10% of the time.
- Unfortunately, the weatherman is forecasting rain for tomorrow.
- What is the probability it will rain on the day of Marie's wedding?

$Y = 1\ (Y)$: *Weatherman forecast it rains*
$H = 1\ (H)$: *It Actually Rains*

Prior Probability

$$P(H) = \frac{5}{365} = 0.0137$$

Likelihood

$$P(Y|H) = 0.9$$

$$P(Y|\widetilde{H}) = 0.1$$

Posterior Probability

$$P(H|Y) = ??$$

© Copyright National University of Singapore. All Rights Reserved.

Bayes' Rule: Example 1 – Weather Forecast

- We know that: $P(H) = \frac{5}{365} = 0.0137$ and $P(Y|H) = 0.9$, $P\left(Y\middle|\widetilde{H}\right) = 0.1$.
  What we want: $P(H|Y)$.

$$P(H|Y) = \frac{P(H)P(Y|H)}{P(Y)} = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + P(Y|\widetilde{H})P\left(\widetilde{H}\right)}$$

$$P(H|Y) = \frac{0.0137(0.9)}{0.9(0.0137) + 0.1(1 - 0.0137)} = 0.111$$

- The probability it will rain on the day of Marie's wedding, given the weatherman is forecasting rain for tomorrow is 0.111

© Copyright National University of Singapore. All Rights Reserved.

Bayes Rule: Example 2 – Disease Testing

- Suppose there is an outbreak of a disease (e.g., COVIDv2). Jack is taking a diagnostic test to determine if he is infected.
- Suppose the sensitivity (true positive rate) of the diagnostic is 87.5%, and the specificity (true negative rate) of the diagnostic is 97.5%.
- Also suppose the prevalence of the disease in the area where Jack lives is 10%.
- Unfortunately, Jack is tested positive. What is the probability of Jack being infected?

$Y = 1\ (Y)$: *Diagnostic test positive*
$H = 1\ (H)$: *Jack is infected*

Likelihood

$$P(Y|H) = 0.875$$
$$P(\tilde{Y}|\tilde{H}) = 0.975$$

Prior Probability

$$P(H) = 0.1$$

Posterior Probability

$$P(H|Y) = ??$$

© Copyright National University of Singapore. All Rights Reserved.

Bayes' Rule: Example 2 – Disease Testing

- We know that: $P(H) = 0.1$ and $P(Y|H) = 0.875$, $P\left(\tilde{Y}\middle|\tilde{H}\right) = 0.975$.
- What we want: $P(H|Y)$.

$$P(H|Y) = \frac{P(H)P(Y|H)}{P(Y)} = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + P(Y|\tilde{H})P(\tilde{H})} = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + \left(1 - P\left(\tilde{Y}\middle|\tilde{H}\right)\right)P(\tilde{H})}$$

$$P(H|Y) = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + \left(1 - P\left(\tilde{Y}\middle|\tilde{H}\right)\right)\left(1 - P(H)\right)} = \frac{0.1 \times 0.875}{0.875 \times 0.1 + (1 - 0.1)(1 - 0.975)} = 0.795$$

- The chance of Jack being infected is 79.5%.

© Copyright National University of Singapore. All Rights Reserved.

Bayes Rule: Example 2 – Disease Testing

- Suppose there is an outbreak of a disease (e.g., COVIDv2). Jack is taking a diagnostic test to determine if he is infected.
- Suppose the sensitivity (true positive rate) of the diagnostic is 87.5%, and the specificity (true negative rate) of the diagnostic is 97.5%.
- Now suppose the prevalence of the disease in the area where Jack lives is changed to 1%.
- Jack is tested positive. What is the probability of Jack being infected now?

$Y = 1\ (Y)$: *Diagnostic test positive*
$H = 1\ (H)$: *Jack is infected*

Likelihood

$$P(Y|H) = 0.875$$
$$P(\tilde{Y}|\tilde{H}) = 0.975$$

Prior Probability

$$P(H) = 0.01$$

Posterior Probability

$$P(H|Y) = ??$$

© Copyright National University of Singapore. All Rights Reserved.

Bayes' Rule: Example 2 – Disease Testing

- We know that: $P(H) = 0.01$ and $P(Y|H) = 0.875$, $P\left(\tilde{Y}\middle|\tilde{H}\right) = 0.975$.
- What we want: $P(H|Y)$.

$$P(H|Y) = \frac{P(H)P(Y|H)}{P(Y)} = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + P(Y|\tilde{H})P\left(\tilde{H}\right)} = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + \left(1 - P\left(\tilde{Y}\middle|\tilde{H}\right)\right)P\left(\tilde{H}\right)}$$

$$P(H|Y) = \frac{P(H)P(Y|H)}{P(Y|H)P(H) + \left(1 - P\left(\tilde{Y}\middle|\tilde{H}\right)\right)\left(1 - P(H)\right)} = \frac{0.01 \times 0.875}{0.875 \times 0.01 + (1 - 0.01)(1 - 0.975)} = 0.261$$

- The chance of Jack being infected is 26.1%, even if tested positive.

© Copyright National University of Singapore. All Rights Reserved.

Bernoulli Distribution

- One of the simplest probability distribution – can be used to model binary events. (E.g., coin flipping)
  - Explaining the Bernoulli distribution with coin flipping. Denote $Y = 1$ as coin lands on heads, and $Y = 0$ as coin lands on tails.
  - Suppose the probability of the coin lands on heads is $0 \leq \theta \leq 1$. Thus, $p(Y = 1) = \theta$ and $p(Y = 0) = 1 - \theta$. Then random variable Y follow the Bernoulli distribution: $Y \sim \mathrm{Ber}(\theta)$.
  - $\mathrm{Ber}(y|\theta) = \begin{cases} 1 - \theta & if\ y = 0 \\ \theta & if\ y = 1 \end{cases} \triangleq \theta^y (1 - \theta)^{1-y}$

© Copyright National University of Singapore. All Rights Reserved.

Binomial Distribution

- Now, if we flip a coin multiple times. Then the sum of the Bernoulli random variables will follow a **Binomial** distribution.
  - The Bernoulli distribution is a **special case** of the binomial distribution.

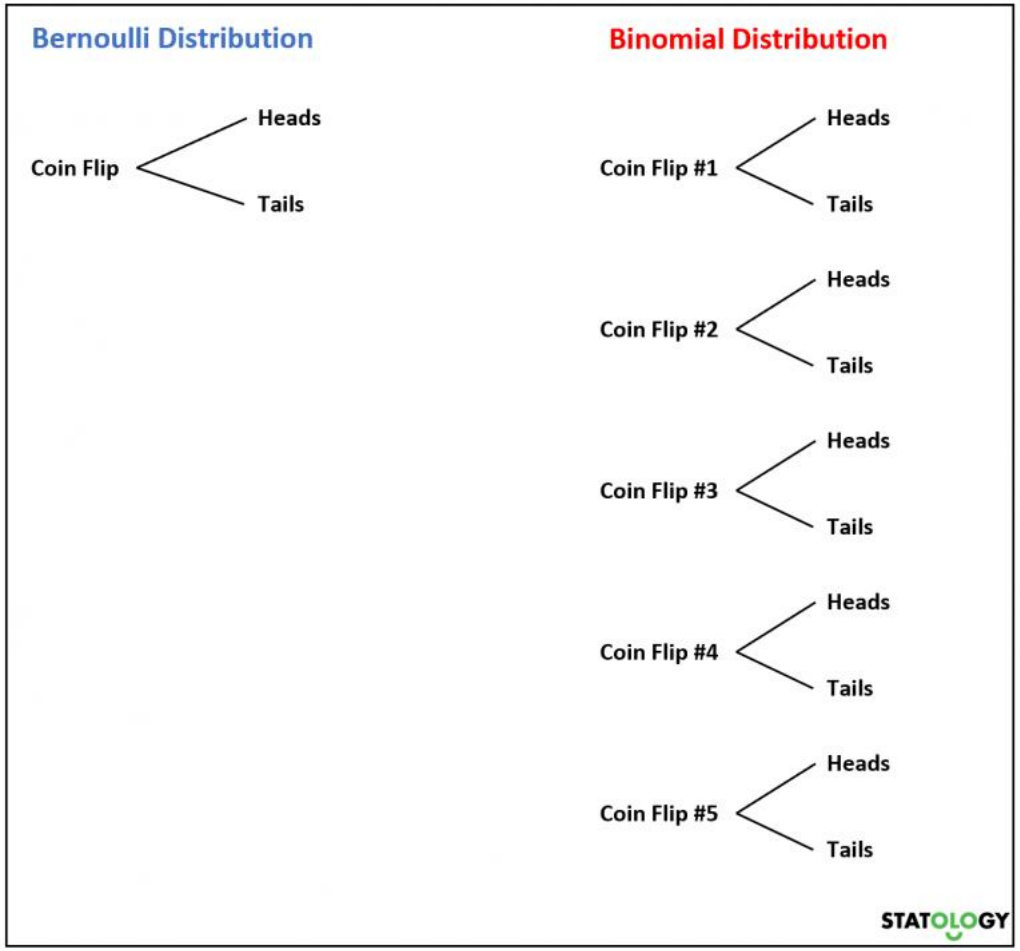
  - Suppose we observe a set of N Bernoulli trials (e.g., toss a coin N times), $y_n \sim \mathrm{Ber}(\cdot|\theta)$ for $n = 1{:}N$. Define s to be the total number of heads $s \triangleq \sum_{n=1}^{N} \mathbb{I}(y_n = 1)$.
  - The distribution of s follows the binomial distribution:

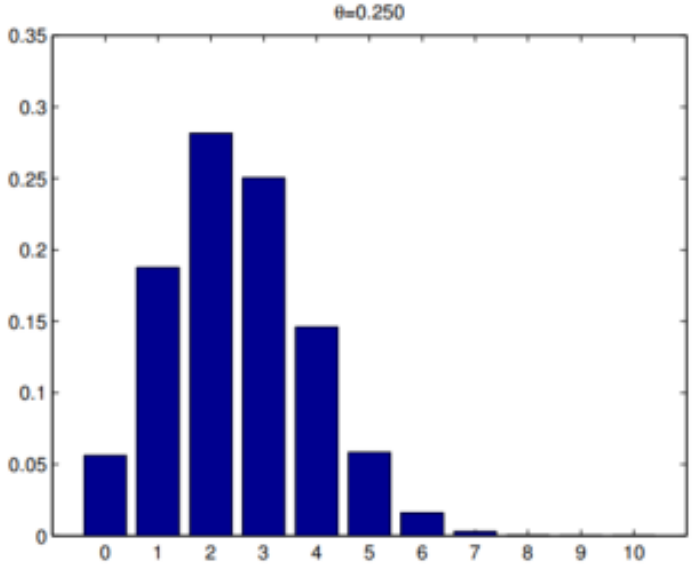
$$\mathrm{Bin}(s|N,\theta) \triangleq \binom{N}{s} \theta^s (1-\theta)^{N-s} = \boldsymbol{C}_s^N \theta^s (1-\theta)^{N-s}$$

$$\text{binomial coefficient: } \binom{N}{s} \triangleq \frac{N!}{(N-s)!\,s!}$$

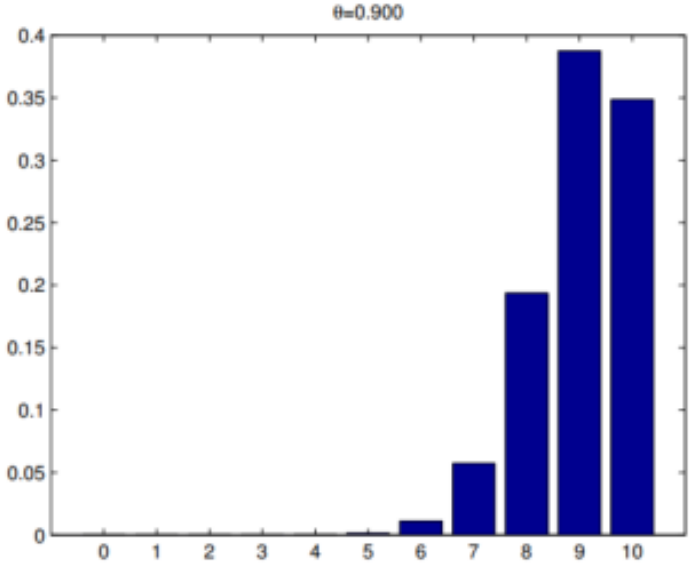
© Copyright National University of Singapore. All Rights Reserved.

Binomial Distribution





(a)



(b)

© Copyright National University of Singapore. All Rights Reserved.

End of Week 1

© Copyright National University of Singapore. All Rights Reserved.

Recap: Bayes' Rule

- "Bayesian": refer to inference methods that represent "degrees of certainty" using probability theory, and which leverage Bayes' rule, to update the degree of certainty given data.
  - *Layman term: Update our belief based on prior belief and data so that we infer something with a certain degree of uncertainty*.
- Bayes' rule: a formula for computing the probability distribution over possible values of an unknown quantity H given observed data $Y = y$.

$$p(H = h|Y = y) = \frac{p(H = h)p(Y = y|H = h)}{p(Y = y)}$$

- Follows the product rule of probability: $p(h|y)p(y) = p(h)p(y|h) = p(h, y)$
- $p(H = h|Y = y)$: **posterior distribution / conditional distribution**.

© Copyright National University of Singapore. All Rights Reserved.

Recap: Bernoulli Distribution

- One of the simplest probability distribution – can be used to model binary events. (E.g., coin flipping)
  - Explaining the Bernoulli distribution with coin flipping. Denote $Y = 1$ as coin lands on heads, and $Y = 0$ as coin lands on tails.
  - Suppose the probability of the coin lands on heads is $0 \leq \theta \leq 1$. Thus, $p(Y = 1) = \theta$ and $p(Y = 0) = 1 - \theta$. Then random variable Y follow the Bernoulli distribution: $Y \sim \mathrm{Ber}(\theta)$.
  - $\mathrm{Ber}(y|\theta) = \begin{cases} 1 - \theta & if\ y = 0 \\ \theta & if\ y = 1 \end{cases} \triangleq \theta^y (1 - \theta)^{1-y}$

© Copyright National University of Singapore. All Rights Reserved.

Recap: Binomial Distribution

- Now, if we flip a coin multiple times. Then the sum of the Bernoulli random variables will follow a **Binomial** distribution.
  - The Bernoulli distribution is a **special case** of the binomial distribution.

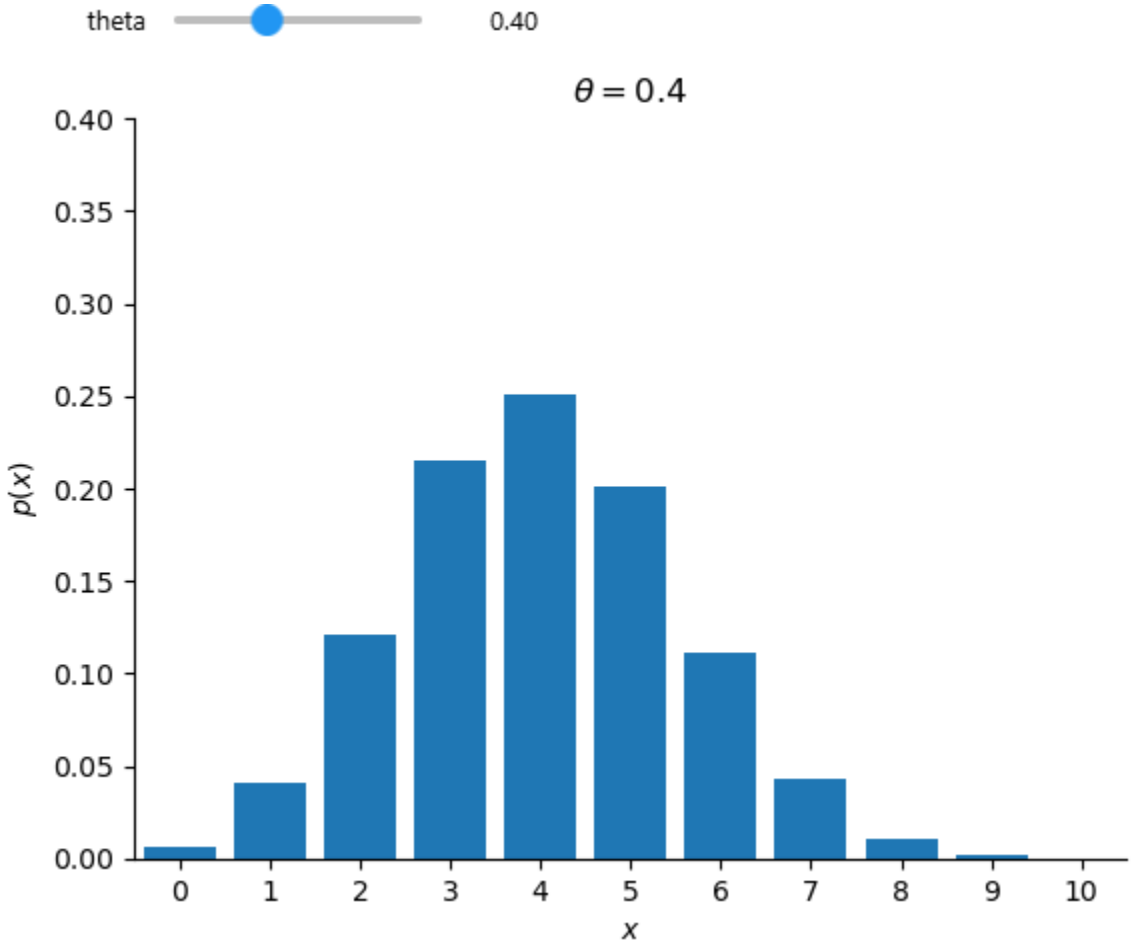
  - Suppose we observe a set of N Bernoulli trials (e.g., toss a coin N times), $y_n \sim \mathrm{Ber}(\cdot|\theta)$ for $n = 1{:}N$. Define s to be the total number of heads $s \triangleq \sum_{n=1}^{N} \mathbb{I}(y_n = 1)$.
  - The distribution of s follows the binomial distribution:

$$\mathrm{Bin}(s|N,\theta) \triangleq \binom{N}{s} \theta^s (1-\theta)^{N-s} = \boldsymbol{C}_s^N \theta^s (1-\theta)^s$$

$$\text{binomial coefficient: } \binom{N}{s} \triangleq \frac{N!}{(N-s)!\,s!}$$

© Copyright National University of Singapore. All Rights Reserved.

Binomial Distribution: Example



© Copyright National University of Singapore. All Rights Reserved.

Predicting a Binary Variable

- When we want to predict a binary variable $y \in \{0,1\}$ given input $x \in \mathcal{X}$, we need to use a conditional probability distribution:
  - $p(y|\boldsymbol{x}, \boldsymbol{\theta}) = \mathrm{Ber}(y|f(\boldsymbol{x}; \boldsymbol{\theta}))$.
  - $f(\boldsymbol{x}; \boldsymbol{\theta})$ is a function that predicts the parameter of the distribution.

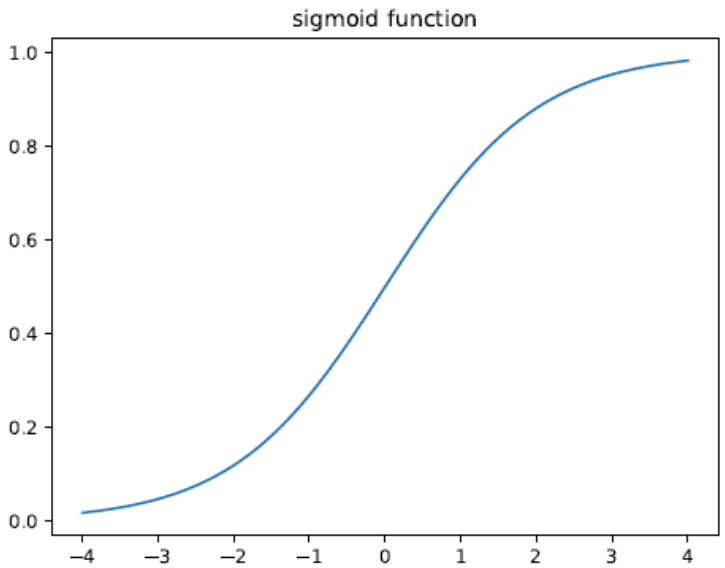
- To avoid requiring $0 < f(\boldsymbol{x}; \boldsymbol{\theta}) \leq 1$, let f be an unconstrained function, we use the following model:
  - $p(y|\boldsymbol{x}, \boldsymbol{\theta}) = \mathrm{Ber}(y|\sigma(f(\boldsymbol{x}; \boldsymbol{\theta})))$.
  - $\sigma()$ is the sigmoid (logistic) function – "sigmoid" means S-shaped.

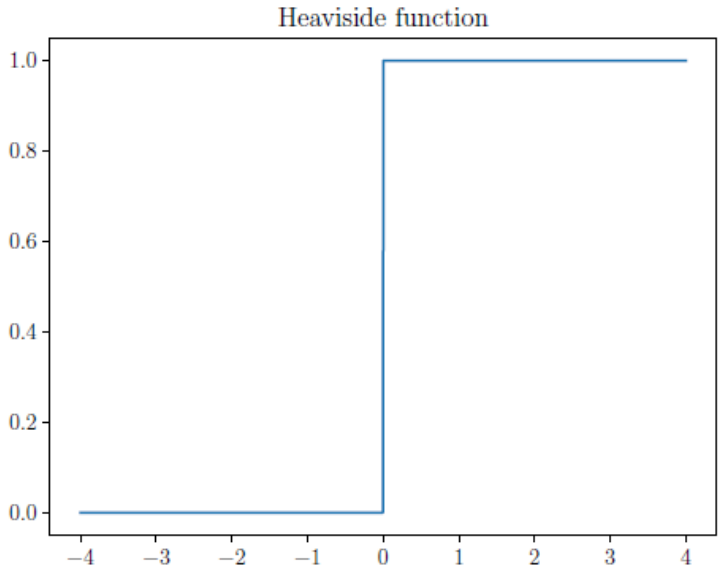
$$\sigma(a) \triangleq \frac{1}{1 + e^{-a}}$$

© Copyright National University of Singapore. All Rights Reserved.

The Sigmoid Function Visualized

- Comparing the sigmoid function $\sigma(a) = (1 + e^{-a})^{-1}$ and the Heaviside step function $\mathbb{I}(a > 0)$.
- The sigmoid function can be thought of as a “soft” version of the heaviside step function.





© Copyright National University of Singapore. All Rights Reserved.

Categorical and Multinomial Distribution

- Problems for Bernoulli and Binomial distributions?
  - Can only model the probability for an event with ONLY 2 outcomes.
  - Generalize the Bernoulli to $C > 2$ values.
- **Categorical distribution**:
  - A discrete probability distribution with one parameter per class:
  - $\mathrm{Cat}(y|\boldsymbol{\theta}) \triangleq \prod_{c=1}^{C} \theta_c^{\mathbb{I}(y=c)}$, thus $p(y = c|\boldsymbol{\theta}) = \theta_c$.
  - Alternatively, convert the discrete variable y into a one-hot vector $\boldsymbol{y}$ with C elements.
  - E.g., if $C = 3$, then the classes are encoded as $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$.
  - As such, the categorical distribution is also written for $\boldsymbol{y}$:

$$\mathrm{Cat}(\boldsymbol{y}|\boldsymbol{\theta}) \triangleq \prod_{c=1}^{C} \theta_c^{y_c}$$

© Copyright National University of Singapore. All Rights Reserved.

Categorical and Multinomial Distribution

- The categorical distribution is a special case of **multinomial distribution**.
  - Repeat the categorical trials N times (e.g., roll a C-sided dice N times).
  - Define y_c to be the total number of times face c shows up: $y_c = N_c \triangleq \sum_{n=1}^{N} \mathbb{I}(y_n = c)$. Define $\boldsymbol{y}$ as the vector with C-dim, each element being y_c.
  - The distribution of $\boldsymbol{y}$ follows the multinominal distribution. Note that $\boldsymbol{y}$ is no longer a one-hot vector.

$$\mathcal{M}(y|N,\theta) \triangleq \binom{N}{y_1 \dots y_c} \prod_{c=1}^{C} \theta_c^{y_c} = \binom{N}{N_1 \dots N_c} \prod_{c=1}^{C} \theta_c^{N_c}$$

  - θ_c is the probability side c shows up and $\binom{N}{N_1 \dots N_c} \triangleq \frac{N!}{N_1! N_2! \dots N_C!}$ is the multinominal coefficient, which is the number of ways to divide a set of size $N = \sum_{c=1}^{C} N_c$ into subsets with sizes N_1 up to N_C. If $N = 1$, the multinomial distribution becomes the categorical distribution, if $C = 2$, the multinomial distribution is the binomial distribution.

© Copyright National University of Singapore. All Rights Reserved.

Predicting a Categorical Variable with Softmax

- When we want to predict a categorical variable y given input $x \in \mathcal{X}$, we need to use a conditional probability distribution:
  - $p(y|\boldsymbol{x}, \boldsymbol{\theta}) = \mathrm{Cat}(y|f(\boldsymbol{x}; \boldsymbol{\theta}))$.
- To avoid requiring $0 < f_c(\boldsymbol{x}; \boldsymbol{\theta}) \leq 1$ and $\sum_{c=1}^{C} f_c(\boldsymbol{x}; \boldsymbol{\theta}) = 1$, let f be an unconstrained function, we use the following model:
  - $p(y|\boldsymbol{x}, \boldsymbol{\theta}) = \mathrm{Cat}(y|\mathrm{softmax}(f(\boldsymbol{x}; \boldsymbol{\theta})))$.
  - $\mathrm{softmax}()$, also written as $\mathrm{sfm}()$ is the softmax (multinomial logistic) function

$$\mathrm{softmax}(\boldsymbol{a})_c \triangleq \left[\frac{e^{a_1}}{\sum_{c'=1}^{C} e^{a_{c'}}}, \dots, \frac{e^{a_C}}{\sum_{c'=1}^{C} e^{a_{c'}}}\right]$$

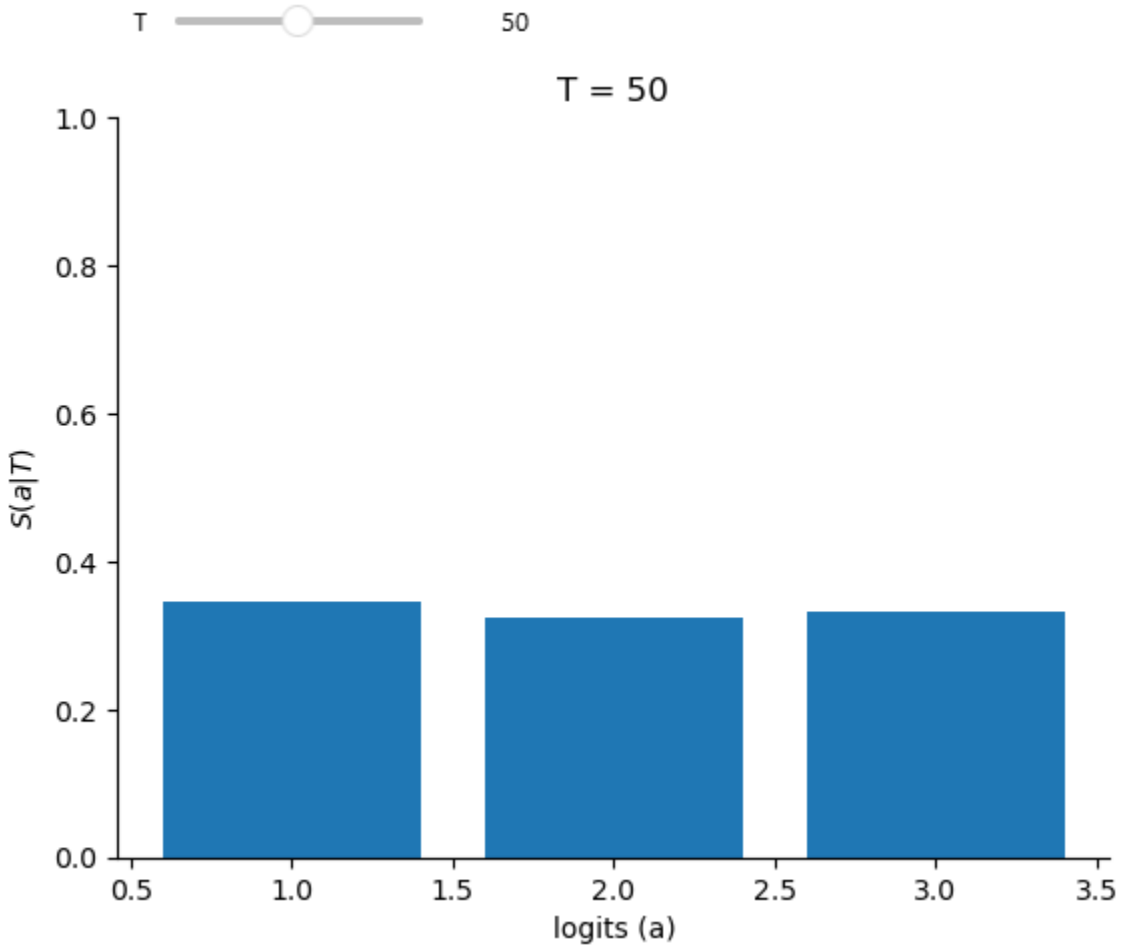
© Copyright National University of Singapore. All Rights Reserved.

Something More on Softmax

- 1. Properties of Softmax:
  - a. $\mathrm{softmax}(\boldsymbol{a})_c \in [0,1]$, and $\sum_{c=1}^{C} \mathrm{softmax}(\boldsymbol{a})_c = 1$. The softmax function has been applied to almost ALL classification tasks in deep learning thanks to these properties.
  - The input to the softmax $\boldsymbol{a} = f(\boldsymbol{x}; \boldsymbol{\theta})$ are called **logits**.
- 2. The softmax function, at its extremes, acts like the argmax function.
  - Divide each a_c (the components for each class of $\boldsymbol{a}$) by a constant T, denoted as the Temperature. Then as $T \to 0$, we find the following:
  - $\mathrm{softmax}(\boldsymbol{a}/T)_c = \begin{cases} 1.0 & \text{if } c = \arg\max_{c'} a_{c'} \\ 0.0 & \text{otherwise} \end{cases}$
  - At low temperatures, the distribution puts most of its probability mass in the most probable state (denoted as **winner takes all**), whereas at high temperatures, it spreads the mass uniformly.

© Copyright National University of Singapore. All Rights Reserved.

Illustration of Softmax Properties



© Copyright National University of Singapore. All Rights Reserved.

Univariate Gaussian Distribution

- The most widely used distribution of real-valued random variables $y \in \mathbb{R}$ is the Gaussian distribution (equivalently, normal distribution).
- Suppose a continuous variable X follow the Gaussian distribution, then cdf is: $P(x) \triangleq P(X \leq x)$ formulated as:

$$\Phi(x; \mu, \sigma^2) \triangleq \int_{-\infty}^{x} \mathcal{N}(z|\mu, \sigma^2) dz = \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{z}{\sqrt{2}}\right)\right]$$

- Here, μ encodes the mean of the distribution, σ^2 encodes the variance, when $\mu = 0$ and $\sigma = 1$, it refers to the **standard normal** distribution.
- $z = (x - \mu)/\sigma$ and $\mathrm{erf}(u)$ is the error function defined as:

$$\mathrm{erf}(u) \triangleq \frac{2}{\sqrt{\pi}} \int_{0}^{u} e^{-t^2} dt$$

© Copyright National University of Singapore. All Rights Reserved.

Univariate Gaussian Distribution

- The pdf is the derivative of the cdf: $p(x) \triangleq \frac{d}{dx} P(x)$ and the pdf of the Gaussian is given by:

$$\mathcal{N}(x|\mu, \sigma^2) \triangleq \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

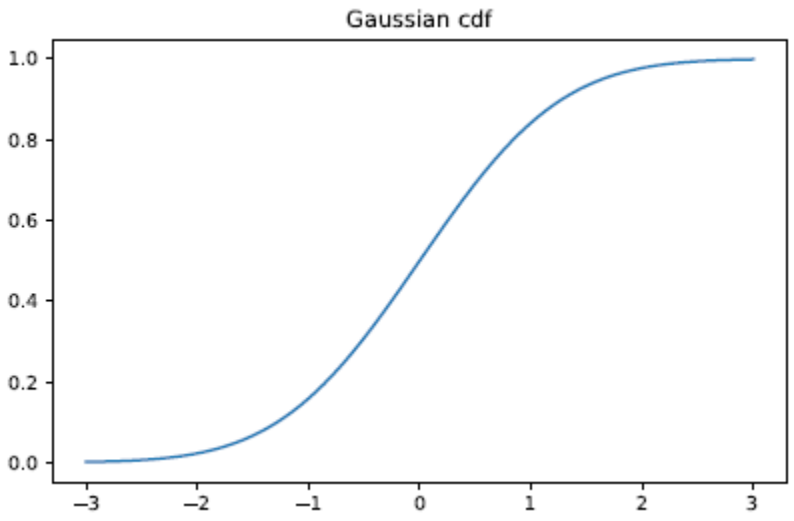
- RV x that follows such pdf can be denoted as: $x \sim \mathcal{N}(\mu, \sigma^2)$.
- For Gaussian, the following results apply:
- Mean (expected value): $\mathbb{E}[\mathcal{N}(\cdot\,|\mu, \sigma^2)] = \mu$.
- Variance: $\mathbb{V}[X] = \mathbb{E}[X^2] - \mu^2$, therefore $\mathbb{E}[X^2] = \mu^2 + \sigma^2$.
- Standard deviation: $\mathrm{std}[X] = \sqrt{\mathbb{V}[X]} = \sigma$.

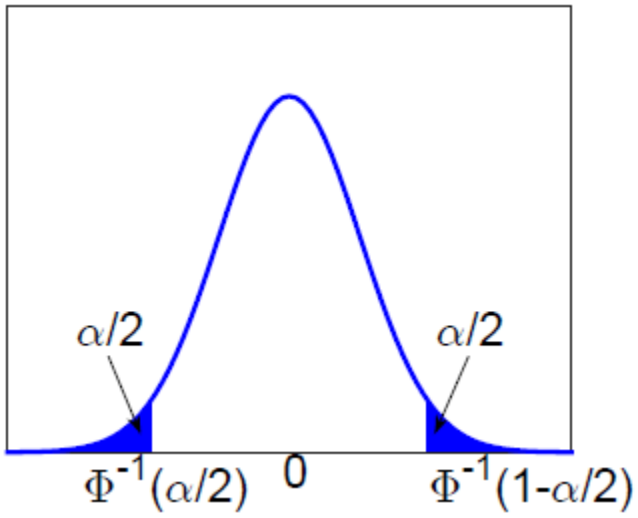
© Copyright National University of Singapore. All Rights Reserved.

Univariate Gaussian Distribution

- The pdf is the derivative of the cdf: $p(x) \triangleq \frac{d}{dx} P(x)$ and the pdf of the Gaussian is given by:

$$\mathcal{N}(x|\mu, \sigma^2) \triangleq \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$



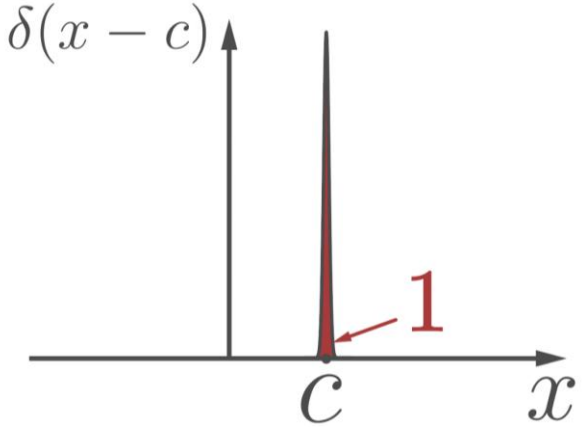


© Copyright National University of Singapore. All Rights Reserved.

Other Common Univariate Distributions

- Dirac delta: when the σ of Gaussian → 0.

$$\lim_{\sigma \to 0} \mathcal{N}(y|\mu, \sigma^2) \to \delta(y-\mu) \quad \delta(x) = \begin{cases} +\infty & \text{if } x = 0 \\ 0 & \text{if } x \neq 0 \end{cases} \quad \int_{-\infty}^{\infty} \delta(x)dx = 1$$

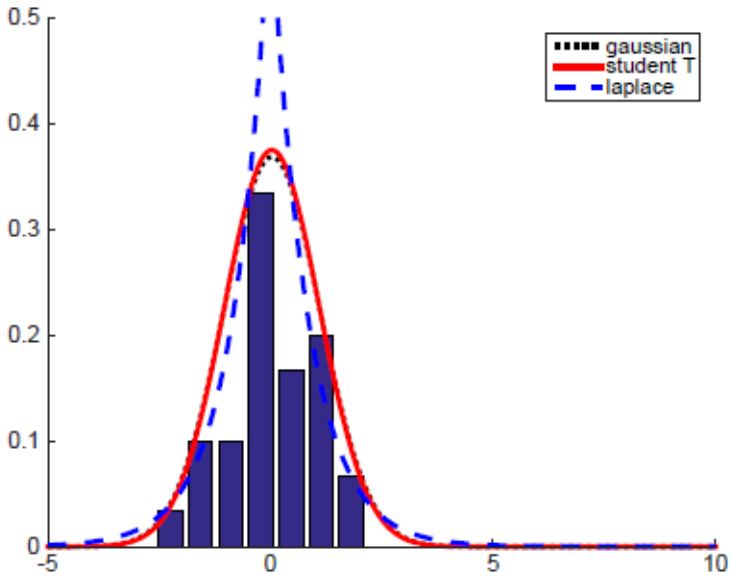


- Student t-distribution: less sensitive to outliers than the Gaussian distribution.

$$\mathcal{T}(y|\mu, \sigma^2, \nu) \propto \left[1 + \frac{1}{\nu}\left(\frac{y-\mu}{\sigma}\right)^2\right]^{-\left(\frac{\nu+1}{2}\right)}$$

- Laplace distribution: double-sided exponential distribution.

$$\text{Laplace}(y|\mu, b) \triangleq \frac{1}{2b}\exp\left(-\frac{|y-\mu|}{b}\right)$$



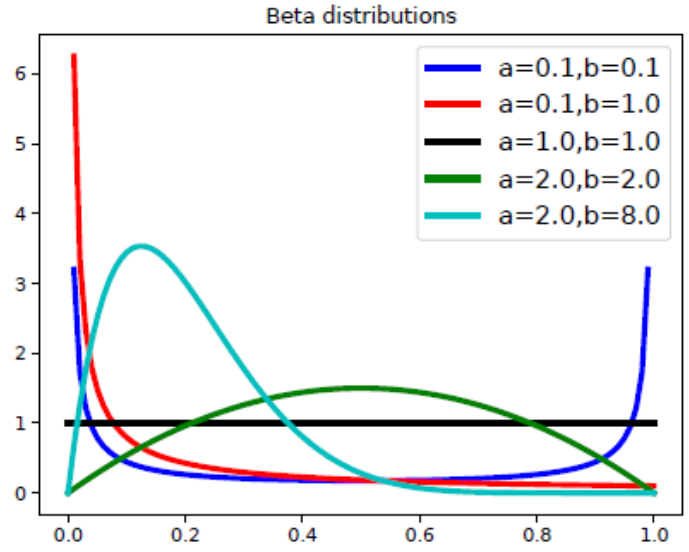
© Copyright National University of Singapore. All Rights Reserved.

Other Common Univariate Distributions

- Beta distribution: a suitable model for the random behavior of percentages and proportions

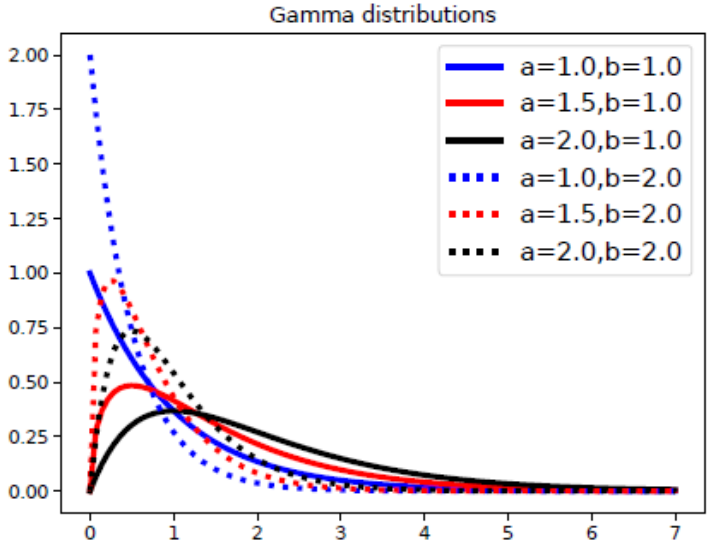
$$\mathrm{Beta}(x|a,b) = \frac{1}{B(a,b)} x^{a-1}(1-x)^{b-1}$$

$$B(a,b) \triangleq \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \quad \Gamma(a) \triangleq \int_0^\infty x^{a-1}e^{-x}dx$$



- Gamma distribution: a flexible distribution for positive real valued rv's.

$$\mathrm{Ga}(x|\mathrm{shape}=a, \mathrm{rate}=b) \triangleq \frac{b^a}{\Gamma(a)} x^{a-1}e^{-xb}$$



© Copyright National University of Singapore. All Rights Reserved.

Estimating Probability Models: Model Fitting

- Now suppose we know that the data would follow a certain probability model, controlled by parameters $\boldsymbol{\theta}$, but we don't know what the model exactly is.
  - Key problem: how to learn these parameters $\boldsymbol{\theta}$ from data?
  - The process of estimation θ from data $\mathcal{D}$ is called **model fitting**, or equivalently, "**training**" – the core of machine learning.
- The training boils down to an optimization of the form:

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$$

  - $\widehat{\boldsymbol{\theta}}$: a point estimate; $\mathcal{L}(\boldsymbol{\theta})$: loss function / objective function.
  - Important: how to define the loss function / objective function?
  - Inference: quantify the uncertainty from data sample – compute $p(y|x, \widehat{\boldsymbol{\theta}})$.

© Copyright National University of Singapore. All Rights Reserved.

Maximum Likelihood Estimation (MLE)

- The intuitive objective: pick the parameters that assign the highest prob. to the training data (usually labelled) – maximum **likelihood** estimation:

$$\widehat{\boldsymbol{\theta}}_{mle} = \arg\max_{\boldsymbol{\theta}} p(\mathcal{D}|\boldsymbol{\theta})$$

  - Assume: training samples are independently sampled from the same distribution – iid assumption (independent and identical distributed).
  - $p(\mathcal{D}|\boldsymbol{\theta}) = \prod_{n=1}^{N} p(y_n|x_n, \boldsymbol{\theta})$ – likelihood; log likelihood takes its log $\ell(\boldsymbol{\theta}) \triangleq \log p(\mathcal{D}|\boldsymbol{\theta})$.
- Most optimization algorithms are designed to min cost functions (argmin instead of argmax), redefine the objective function to be the (conditional) negative log likelihood (NLL): maximize likelihood = minimize NLL.

$$\mathrm{NLL}(\boldsymbol{\theta}) \triangleq -\log p(\mathcal{D}|\boldsymbol{\theta}) = -\sum_{n=1}^{N} \log p(y_n|x_n, \boldsymbol{\theta})$$

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) = \arg\min_{\boldsymbol{\theta}} \mathrm{NLL}(\boldsymbol{\theta})$$

© Copyright National University of Singapore. All Rights Reserved.

MLE for Bernoulli Distribution

- Suppose Y is a random variable representing a coin toss, where the event $Y = 1$ corresponds to heads and $Y = 0$ corresponds to tails.
- Let $\theta = p(Y = 1)$ be the probability of heads. The probability distribution for this rv is the Bernoulli. The NLL for the Bernoulli distribution is given by:
  - Observe: this is just the empirical fraction of heads, an intuitive result!

$$
\begin{aligned}
\mathrm{NLL}(\theta) &= -\log \prod_{n=1}^{N} p(y_n|\theta) \\
&= -\log \prod_{n=1}^{N} \theta^{\mathbb{I}(y_n=1)}(1-\theta)^{\mathbb{I}(y_n=0)} \\
&= -\sum_{n=1}^{N} \left[\mathbb{I}\left(y_n = 1\right)\log\theta + \mathbb{I}\left(y_n = 0\right)\log(1-\theta)\right] \\
&= -[N_1 \log\theta + N_0 \log(1-\theta)]
\end{aligned}
$$

The MLE can be found by solving $\frac{d}{d\theta}\mathrm{NLL}(\theta) = 0$.

$$
\frac{d}{d\theta}\mathrm{NLL}(\theta) = \frac{-N_1}{\theta} + \frac{N_0}{1-\theta}
$$

and hence the MLE is given by

$$
\hat{\theta}_{\mathrm{mle}} = \frac{N_1}{N_0 + N_1}
$$

© Copyright National University of Singapore. All Rights Reserved.

MLE for Univariate Gaussian Distribution

- Suppose $Y \sim N(\mu, \sigma^2)$ and $\mathcal{D} = \{y_n : n = 1 : N\}$ is an iid sample of size N. The parameters are $\boldsymbol{\theta} = (\mu, \sigma^2)$. We estimate with MLE:

- 1. Derive NLL:

$$\mathrm{NLL}(\mu, \sigma^2)$$

$$= -\sum_{n=1}^{N} \log\left[\left(\frac{1}{2\pi\sigma^2}\right)^{\frac{1}{2}} \exp\left(-\frac{1}{2\sigma^2}(y_n - \mu)^2\right)\right]$$

$$= \frac{1}{2\sigma^2}\sum_{n=1}^{N}(y_n - \mu)^2 + \frac{N}{2}\log(2\pi\sigma^2)$$

- 2. Derive optimal parameter:

$$\frac{\partial}{\partial \mu}\mathrm{NLL}(\mu, \sigma^2) = 0, \quad \frac{\partial}{\partial \sigma^2}\mathrm{NLL}(\mu, \sigma^2) = 0$$

$$\hat{\mu}_{\mathrm{mle}} = \frac{1}{N}\sum_{n=1}^{N} y_n = \overline{y}$$

$$s^2 \triangleq \frac{1}{N}\sum_{n=1}^{N} y_n^2$$

$$\hat{\sigma}^2_{\mathrm{mle}} = \frac{1}{N}\sum_{n=1}^{N}(y_n - \hat{\mu}_{\mathrm{mle}})^2$$

$$= \frac{1}{N}\sum_{n=1}^{N}\left[y_n^2 + \hat{\mu}^2_{\mathrm{mle}} - 2y_n\hat{\mu}_{\mathrm{mle}}\right] = s^2 - \overline{y}^2$$

© Copyright National University of Singapore. All Rights Reserved.

Empirical Risk Minimization (ERM)

- The MLE is based on: $\mathrm{NLL}(\boldsymbol{\theta}) = -\sum_{n=1}^{N} \log p(y_n | x_n, \boldsymbol{\theta})$
  - Can we further generalize the MLE?
  - Recall: get optimal $\boldsymbol{\theta}$ by $\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) = \arg\min_{\boldsymbol{\theta}} \sum_{n=1}^{N} \ell(\boldsymbol{\theta})$.
- Generalize MLE: replace log loss term with other loss function!
- $\ell(\boldsymbol{\theta}) = -\log p(y_n | x_n, \boldsymbol{\theta}) \rightarrow \ell(y_n, \boldsymbol{\theta}; x_n)$.
- With the more generalized form, we have:
- $\mathcal{L}(\boldsymbol{\theta}) = \sum_{n=1}^{N} \ell(y_n, \boldsymbol{\theta}; x_n)$ – Empirical risk minimization (ERM).
  - The expected loss where the expectation is taken w.r.t. the “empirical” (sampled) distribution.

© Copyright National University of Singapore. All Rights Reserved.

ERM: Minimizing Misclassification Rate

- If we are solving a classification problem, an intuitive loss selection would be the 0-1 loss: (f being some kind of predictor)

$$\ell_{01}(y_n, \boldsymbol{\theta}; x_n) = \begin{cases} 0 & if\ y_n = f(x_n; \boldsymbol{\theta}) \\ 1 & if\ y_n \neq f(x_n; \boldsymbol{\theta}) \end{cases}$$

- Then the empirical risk becomes:

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{n=1}^{N} \ell_{01}(y_n, \boldsymbol{\theta}; x_n)$$

- This is defined as the empirical **misclassification rate** on the training set.

© Copyright National University of Singapore. All Rights Reserved.